Name _____ Date _____ Period _____

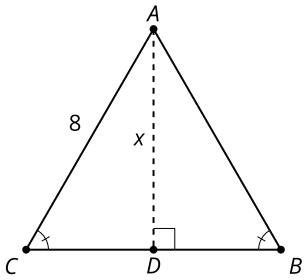
# Right Triangles and Trigonometry

Geometry · Right Triangle Trigonometry

**1.** Triangle $ABC$ is equilateral.

a. What is the value of $x$?

b. What is the measure of angle $B$?



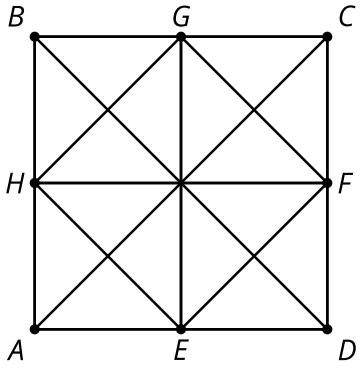

**2.** An equilateral triangle has side length 8 units. What is the area?

**3.** The quilt is made of squares with diagonals.

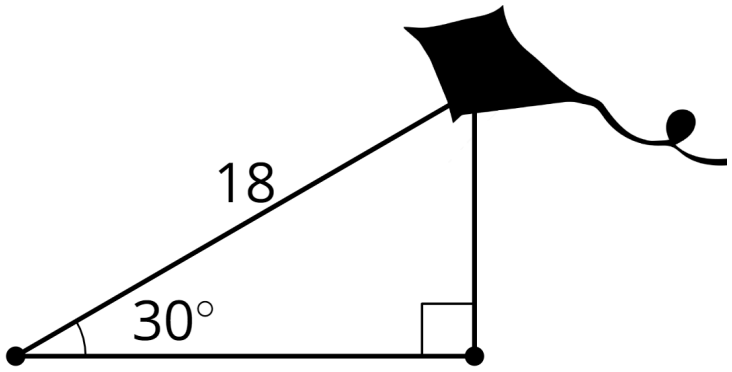
The length of $BD$ is 4.

a. Find the length of $AE$.

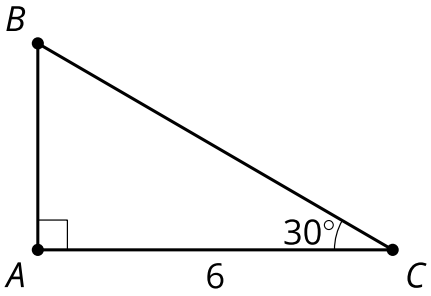
b. Find the area of square $ABCD$.



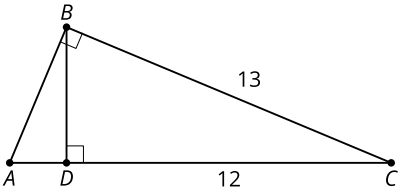
**4.** Kiran is flying a kite. He gets tired, so he stakes the kite into the ground. The kite is on a string that is 18 feet long and makes a 30 degree angle with the ground. How high is the kite?



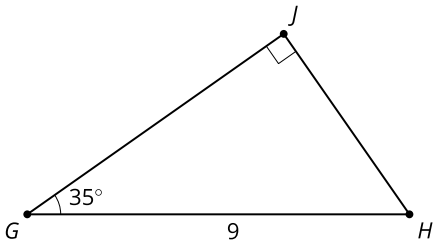

**5.** What is the length of side $AB$?




**6.** Find the lengths of segments $AD$ and $BD$. Then check your answers using a different method.



**7.** Write an expression that can be used to find the length of $JH$ and an expression that can be used to find the length of $GJ$.

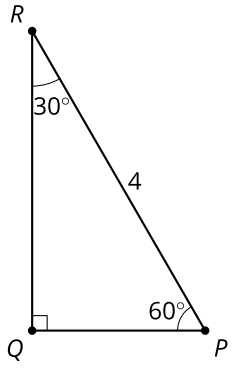



**8.** A triangle has sides with lengths 5, 12, and 13.

a. Verify this is a Pythagorean triple.

b. Approximate the acute angles in this triangle.

**9.** Find the length of each leg.




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